

KKT-free Constrained Optimization and Application to River-Flow Modeling ^{*}

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Abstract

This paper explores Augmented Lagrangian methods for derivative-free continuous optimization with constraints. Our theoretical results do not assume the existence of derivatives, and KKT conditions are not employed. We present numerical experiments on fitting parameters of river model flows.

Key words: River flow modeling, Augmented Lagrangian, convergence, derivative-free optimization.

1 Introduction

Constrained optimization algorithms that do not use derivatives have been a fruitful area of research, driven by their applicability to a wide range of problems in engineering and applied sciences. See [37] for a recent application to fitting parameters of Climate Models. Augmented Lagrangian methods without derivatives of the objective function and constraints have been extensively analyzed in the literature. See [4, 5, 23, 30, 35, 36, 43], among others. Most Augmented Lagrangian algorithms involve solving a single constrained (typically box-constrained) problem at each iteration. Consequently, their efficiency is closely tied to the efficiency of single-constraint derivative-free algorithms [2, 3, 5, 17, 12, 13, 22, 30, 31, 33, 38, 39, 40, 48, 49].

In this paper, we propose a general framework for the Augmented Lagrangian approach, where the choice of the “approximate solution” of the subproblems is arbitrary. From a theoretical perspective, we examine the properties of the resulting algorithm under this general assumption. Our approach differs from most Derivative-Free Augmented Lagrangian methods in the literature, which, although avoiding derivatives in computations, typically assume their existence and continuity to establish KKT-like convergence properties.

The numerical experiments presented in this paper focus on estimating parameters for river-flow models, motivated by the activities of the CRIAB group. This group, established in 2019 at the University of Campinas, aims to develop mathematical models for natural disaster prevention, in response to the devastating collapse of the Brumadinho dam in Minas Gerais, Brazil. The CRIAB group comprises members from various Latin American universities and institutions.

In addition to addressing a critical problem in natural disaster prevention, our experiments also serve as a testbed for the new implementation of Powell’s software BOBYQA [48], recently introduced in [49]. This implementation came to our attention following a presentation by Z. Zhang at the ICIAM Conference in Tokyo, 2023.

This paper is organized as follows. In Section 2 we present the general algorithm and we prove the theoretical results. In Section 3 we describe the problem of fitting parameters for models of natural

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channels. In Section 4 we report the numerical results that correspond to this problem. In Section 5 we present conclusions and lines for future research.

Notation

- The symbol $\|\cdot\|$ denotes an arbitrary vector norm.
- $K_1 \subset K_2$ means that both K_1 and K_2 are sequences of natural numbers and K_1 is a subsequence of K_2 .
- For all $y \in \mathbb{R}^n$, $y_+ = (\max\{0, y_1\}, \dots, \max\{0, y_n\})^T$.
- $\mathbb{N} = \{0, 1, 2, \dots\}$.

2 Derivative-free Safeguarded Augmented Lagrangian

The form of the problems addressed in this paper is as follows:

$$\text{Minimize } f(x); \text{ subject to } h(x) = 0, g(x) \leq 0, x \in \Omega, \quad (1)$$

where $\Omega \subset \mathbb{R}^n$ is a compact set. We assume that the functions $f : \mathbb{R}^n \rightarrow \mathbb{R}, h : \mathbb{R}^n \rightarrow \mathbb{R}^m, g : \mathbb{R}^n \rightarrow \mathbb{R}^p$ are continuous. First derivatives are not assumed to exist anywhere.

Given $\rho > 0$, $\lambda \in \mathbb{R}^m$, $\mu \in \mathbb{R}_+^p$, $x \in \mathbb{R}^n$, we define the Augmented Lagrangian $L_\rho(x, \lambda, \mu)$ by:

$$L_\rho(x, \lambda, \mu) = f(x) + \frac{\rho}{2} \left\{ \sum_{i=1}^m \left[h_i(x) + \frac{\lambda_i}{\rho} \right]^2 + \sum_{i=1}^p \left[g_i(x) + \frac{\mu_i}{\rho} \right]_+^2 \right\}. \quad (2)$$

The definition (2) corresponds to the well-known PHR (Powell-Hestenes-Rockafellar) [29, 47, 52] classical Augmented Lagrangian formulation. See, also, [10, 11, 15, 16].

The idea is that the best strategy to minimize a function subject to constraints may not be to minimize the objective function plus a large penalty for constraint violations, but rather to minimize the objective function plus a moderate penalty for suitably shifted constraints. By properly choosing the shifts λ_i/ρ and μ_i/ρ , it is likely that very large penalty parameters ρ will not be necessary. Notably, the plausibility of this idea does not depend on the differentiability of the objective function or the constraints.

Algorithm 2.1 is a minimalist variant of the Safeguarded Augmented Lagrangian method [11]. The way in which $x^k \in \Omega$ is computed at every iteration of the method is not specified at all. In practice, x^k will be usually obtained by means of some minimization criterion with respect to the augmented Lagrangian.

Algorithm 2.1

Let $0 < \tau < 1$, $\gamma > 1$, $\lambda_{\min} < \lambda_{\max}$, $\mu_{\max} > 0$. At the first outer iteration we use a penalty parameter $\rho_1 > 0$ and safeguarded Lagrange multipliers estimates $\lambda^1 \in \mathbb{R}^m$ and $\mu^1 \in \mathbb{R}^p$ such that

$$\lambda_i^1 \in [\lambda_{\min}, \lambda_{\max}] \quad \forall i = 1, \dots, m \quad \text{and} \quad \mu_i^1 \in [0, \mu_{\max}] \quad \forall i = 1, \dots, p.$$

Step 1. Initialization.

Set $k \leftarrow 1$.

Step 2. Compute the new iterate

Define $\lambda^k \in [\lambda_{\min}, \lambda_{\max}]^m$ and $\mu^k \in [0, \mu_{\max}]^p$.

Compute $x^k \in \Omega$.

Step 3. Update penalty parameter

Define, for all $i = 1, \dots, p$,

$$V_i^k = \max \left\{ g_i(x^k), -\frac{\mu_i^k}{\rho_k} \right\},$$

If $k > 1$ and

$$\max\{\|h(x^k)\|_\infty, \|V^k\|_\infty\} \leq \tau \max\{\|h(x^{k-1})\|_\infty, \|V^{k-1}\|_\infty\}, \quad (3)$$

define

$$\rho_{k+1} = \rho_k.$$

Else, define

$$\rho_{k+1} = \gamma \rho_k.$$

Set $k \leftarrow k + 1$ and go to Step 2.

For all $k \in \mathbb{N}$ we define $N_k = N(x^k)$ as the set of $z \in \Omega$ such that

$$L_{\rho_k}(x^k, \lambda^k, \mu^k) \leq L_{\rho_k}(z, \lambda^k, \mu^k). \quad (4)$$

The points of N_k will be called “neighbors” of x^k . By extension, if x^* is a limit point of $\{x^k\}$, the set of neighbors $N(x^*)$ will be defined by

$$z \in N(x^*) \text{ if and only if there exists } K \subseteq \mathbb{N} \text{ and } z^k \in N_k \text{ such that } \lim_{k \in K} z^k = z.$$

Remarks

- Note that we do not assume that N_k is computed by the algorithm. In the case that we compute the global minimizer of $L_{\rho_k}(x, \lambda^k, \mu^k)$ we have that $N_k = \Omega$.
- Note that, in Algorithm 2.1, the set Ω is quite general, accommodating complicated nonlinear constraints and even integer constraints. Consequently, the computation of x^k can be performed using alternative constrained Derivative-Free algorithms, such as those introduced in [5, 12, 13, 24, 26, 38, 39, 46].

In the following theorem we prove that if $\{x^k\}$ converges to x^* and $\{\rho_k\}$ is bounded, x^* is a feasible point. If ρ_k is not bounded and a sequence of points $z^k \in N_k$ converges to z then x^* is at least as feasible as z .

Therefore, if x^k is a global minimizer of $L_{\rho_k}(x, \lambda^k, \mu^k)$ (so, $N_k = \Omega$), it turns out that x^* is a global minimizer of the sum of squares of infeasibilities. In this sense, Theorem 2.1 is a generalization of Theorem 5.1 of [11].

Theorem 2.1 Assume that $\{x^k\}$ is a sequence generated by Algorithm 2.1. Let $K \subseteq \mathbb{N}$ be such that

$$\lim_{k \in K} x^k = x^*. \quad (5)$$

Then, if $\{\rho_k\}$ is bounded, we have that $h(x^*) = 0$ and $g(x^*) \leq 0$.

Otherwise, if a sequence $\{z^k\} \subset \Omega$ exists such that

$$\lim_{k \in K} z^k = z \quad (6)$$

and $z^k \in N_k$ for all $k \in K$, we have that $z \in \Omega$ and

$$\|h(x^*)\|_2^2 + \|g(x^*)_+\|_2^2 \leq \|h(z)\|_2^2 + \|g(z)_+\|_2^2. \quad (7)$$

Proof Since Ω is compact, by (5), we have that $x^* \in \Omega$.

Assume, firstly, that $\{\rho_k\}$ is bounded. By Step 4, there exists k_0 such that $\rho_k = \rho_{k_0}$ for all $k \geq k_0$. Therefore,

$$\max\{\|h(x^k)\|_\infty, \|V^k\|_\infty\} \leq \tau \max\{\|h(x^{k-1})\|_\infty, \|V^{k-1}\|_\infty\}$$

for all $k \geq k_0$. Then,

$$\lim_{k \rightarrow \infty} \|h(x^k)\| = \lim_{k \rightarrow \infty} \|V^k\| = 0. \quad (8)$$

So, for all $i = 1, \dots, p$,

$$\lim_{k \rightarrow \infty} \left| \max \left\{ g_i(x^k), -\frac{\mu_i^k}{\rho_k} \right\} \right| = 0.$$

Since $\mu_i^k \geq 0$ this implies that

$$\lim_{k \rightarrow \infty} g_i(x^k)_+ = 0. \quad (9)$$

By (8) and (9) we have that $\|h(x^*)\| = 0$ and $\|g(x^*)_+\| = 0$.

Now suppose that $\{\rho_k\}$ is unbounded. By compactness, $z \in \Omega$.

By (4), for all $k \in K$ we have that

$$L_{\rho_k}(x^k, \lambda^k, \mu^k) \leq L_{\rho_k}(z^k, \lambda^k, \mu^k). \quad (10)$$

Then, by the definition of the Augmented Lagrangian, for all $k \in K$ we have that

$$f(x^k) + \frac{\rho_k}{2} \left\{ \sum_{i=1}^m \left[h_i(x^k) + \frac{\lambda_i^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(x^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\} \leq f(z^k) + \frac{\rho_k}{2} \left\{ \sum_{i=1}^m \left[h_i(z^k) + \frac{\lambda_i^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(z^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\}.$$

So,

$$\frac{1}{\rho_k} f(x^k) + \frac{1}{2} \left\{ \sum_{i=1}^m \left[h_i(x^k) + \frac{\lambda_i^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(x^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\} \leq \frac{1}{\rho_k} f(z^k) + \frac{1}{2} \left\{ \sum_{i=1}^m \left[h_i(z^k) + \frac{\lambda_i^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(z^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\}.$$

Taking limits for $k \in K$ and using that λ^k and μ^k are bounded, $\rho_k \rightarrow \infty$, and the continuity of h , g , and $\max$, we obtain:

$$\sum_{i=1}^m h_i(x^*)^2 + \sum_{i=1}^p g_i(x^*)_+^2 \leq \sum_{i=1}^m h_i(z)^2 + \sum_{i=1}^p g_i(z)_+^2.$$

So, (7) is proved. $\square$

Note that for proving this theorem it is essential that the shifts tend to zero when the penalty parameter tends to infinity. This property aligns with common sense, as imposing extreme penalization for infeasibility would render the introduction of shifts unnecessary.

Theorem 2.1 indicates that the likelihood of finding feasible points improves when the penalty parameters are bounded. However, the boundedness of these parameters is not under direct control, as their potential increase at each iteration is determined by the test (3). Properly updating the shifts λ^k/ρ_k and μ^k/ρ_k increases the chances of fulfilling of (3). The formulae

$$\lambda^{k+1} = \lambda^k + \rho_k h(x^k) \text{ and } \mu^{k+1} = (\mu^k + \rho_k g(x^k))_+, \quad (11)$$

subject to the safeguarded defined by $\lambda_{min}, \lambda_{max}, \mu_{max}$, are commonly used. A justification of these updates, without gradient-like arguments, may be found in [11] (Section 4.1). This update strategy plays a crucial role in balancing the penalty parameters and promoting convergence to feasible points.

Nevertheless, it is impossible to prove that the algorithm finds feasible points, as such points may not exist. Therefore, it is sensible to analyze the behavior of the algorithm's limit points, assuming feasibility. The following theorem explores this scenario. In this theorem we will assume boundedness of the penalty

parameters, which, according to Theorem 2.1, is the sufficient condition that guarantees feasibility of any limit point.

Theorem 2.2 Assume that $\{x^k\}$ is a sequence generated by Algorithm 2.1 and that $\{\rho_k\}$ is bounded. Let $K \subseteq \mathbb{N}$ be such that

$$\lim_{k \in K} x^k = x^*. \quad (12)$$

Suppose that, for all $k \in K$, $z^k \in N_k$ is such that

$$\lim_{k \in K} z^k = z, \quad (13)$$

where

$$h(z) = 0 \text{ and } g(z) \leq 0. \quad (14)$$

Then, $h(x^*) = 0, g(x^*) \leq 0$ and

$$f(x^*) \leq f(z).$$

Proof By Theorem 2.1, since $\{\rho_k\}$ is bounded, we have that

$$h(x^*) = 0 \text{ and } g(x^*) \leq 0.$$

By the definition of the Augmented Lagrangian, for all $k \in K$ we have that

$$f(x^k) + \frac{\rho_k}{2} \left\{ \sum_{i=1}^m \left[h_i(x^k) + \frac{\lambda_i^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(x^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\} \leq f(z^k) + \frac{\rho_k}{2} \left\{ \sum_{i=1}^m \left[h_i(z^k) + \frac{\lambda_i^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(z^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\}.$$

Let $K_1 \subseteq K$ be such that $\rho_k \rightarrow \rho, \lambda^k \rightarrow \lambda^*, \mu^k \rightarrow \mu^*$ for $k \in K_1$. Then,

$$f(x^*) + \frac{\rho}{2} \left\{ \sum_{i=1}^m \left[h_i(x^*) + \frac{\lambda_i^*}{\rho} \right]^2 + \sum_{i=1}^p \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\} \leq f(z) + \frac{\rho}{2} \left\{ \sum_{i=1}^m \left[h_i(z) + \frac{\lambda_i^*}{\rho} \right]^2 + \sum_{i=1}^p \left[g_i(z) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\}.$$

Then,

$$f(x^*) + \frac{\rho}{2} \left\{ \sum_{i=1}^m \left[\frac{\lambda_i^*}{\rho} \right]^2 + \sum_{i=1}^p \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\} \leq f(z) + \frac{\rho}{2} \left\{ \sum_{i=1}^m \left[\frac{\lambda_i^*}{\rho} \right]^2 + \sum_{i=1}^p \left[g_i(z) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\}.$$

So,

$$f(x^*) + \frac{\rho}{2} \left\{ \sum_{i=1}^p \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\} \leq f(z) + \frac{\rho}{2} \left\{ \sum_{i=1}^p \left[g_i(z) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\}.$$

Thus,

$$f(x^*) + \frac{\rho}{2} \left\{ \sum_{g_i(x^*) < 0} \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\} + \sum_{g_i(x^*) = 0} \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 \leq f(z) + \frac{\rho}{2} \sum_{i=1}^p \left[\frac{\mu_i^*}{\rho} \right]^2.$$

If $g_i(x^*) = 0$ we have that $[g_i(x^*) + \mu_i^*/\rho]_+ = \mu_i^*/\rho$, then

$$f(x^*) + \frac{\rho}{2} \left\{ \sum_{g_i(x^*) < 0} \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\} \leq f(z) + \frac{\rho}{2} \sum_{g_i(x^*) < 0} \left[\frac{\mu_i^*}{\rho} \right]^2. \quad (15)$$

Finally, if $g_i(x^*) < 0$, by the continuity of g_i we have that $g_i(x^k) < -c = g_i(x^*)/2$ for $k \in K$ large enough. But, by the boundedness of $\{\rho_k\}$ we have that $V_k \rightarrow 0$. Then, by the definition of V_k , $\mu_i^k/\rho_k \rightarrow 0$ for $k \in K$ large enough. So, $\mu_i^*/\rho = 0$. Then, by (15),

$$f(x^*) \leq f(x^*) + \frac{\rho}{2} \left\{ \sum_{g_i(x^*) < 0} \left[g_i(x^*) + \frac{\mu_i^*}{\rho} \right]_+^2 \right\} \leq f(z),$$

as we wanted to prove. $\square$

For completeness, we prove a theorem that concerns optimality in the case that $\rho_k \rightarrow \infty$.

Theorem 2.3 *Assume that $\{x^k\}$ is generated by Algorithm 2.1 and with the hypotheses (12,13,14) and $\rho_k \rightarrow \infty$. In addition, assume that*

$$\lim_{k \in K} \|h(z^k)\| \sqrt{\rho_k} = 0, \text{ and } \lim_{k \in K} \|g(z^k)_+\| \sqrt{\rho_k} = 0. \quad (16)$$

Then, $f(x^*) \leq f(z)$.

Proof Note that, since $z^k = N(x^k)$,

$$\begin{aligned} f(x^k) &+ \frac{\rho_k}{2} \left\{ \sum_{i=1}^m \left[h_i(x^k) + \frac{\lambda_i^k}{\rho_k} \right]^2 + \sum_{i=1}^p \left[g_i(x^k) + \frac{\mu_i^k}{\rho_k} \right]_+^2 \right\} \\ &\leq f(z^k) + \frac{1}{2} \left\{ \sum_{i=1}^m \left[\sqrt{\rho_k} h_i(z^k) + \frac{\lambda_i^k}{\sqrt{\rho_k}} \right]^2 + \sum_{i=1}^p \left[\sqrt{\rho_k} g_i(z^k) + \frac{\mu_i^k}{\sqrt{\rho_k}} \right]_+^2 \right\}. \end{aligned}$$

Then,

$$f(x^k) \leq f(z^k) + \frac{1}{2} \left\{ \sum_{i=1}^m \left[\sqrt{\rho_k} h_i(z^k) + \frac{\lambda_i^k}{\sqrt{\rho_k}} \right]^2 + \sum_{i=1}^p \left[\sqrt{\rho_k} g_i(z^k) + \frac{\mu_i^k}{\sqrt{\rho_k}} \right]_+^2 \right\}. \quad (17)$$

Taking limits in both sides of (17) we obtain the desired result. $\square$

Let us examine the implications of Theorems 2.1 and 2.2. Clearly, the behavior of $\{\rho_k\}$ plays an essential role. In fact, by Theorem 2.1, the boundedness of $\{\rho_k\}$ guarantees that a feasible point is found. This occurs when a value of ρ_k is reached beyond which merely shift modifications are enough to guarantee feasibility. Of course, this favorable situation is impossible if the problem (1) is infeasible. On the other hand, if (1) is feasible, penalty boundedness usually takes place when updates (11) are employed and x^k is computed by means of an adequate procedure.

If ρ_k tends to infinity the thesis of Theorem 2.1 is weaker, as it only says that the limit point x^* is at least as feasible as its neighbors. Nevertheless, this does not preclude x^* from being a feasible point.

For these reasons, since our goal is to find minimizers of (1), we place particular emphasis on Theorem 2.2. This theorem specifies what happens to the objective function when the sufficient condition for feasibility is met. Specifically, Theorem 2.2 states that the objective function value at a limit point is less than or equal to the objective function value at its feasible neighbors. To maximize the impact of this result, it is desirable that the inequality $f(z) \leq f(x^*)$ holds for as many points as possible. A natural strategy, therefore, is to design algorithms where the cardinality of $N(z^k)$ is as large as possible. Although this may require excessive computational time, it is worth noting that the set $N(x^k)$ does not need to be explicitly computed.

From a practical perspective, it is recommendable to include constraints of the form $f(x) \leq c$ in the original problem. By doing so, the effects of Theorem 2.1 can be beneficial in terms of optimality.

3 Application to the estimation of Manning coefficients

The Saint-Venant equations [14, 32, 44] describe the flow of fluids in natural channels. The state variables that describe a natural channel vary with respect to $x \in [x_{min}, x_{max}]$ (position, usually measured in meters) and $t \in [t_{min}, t_{max}]$ (time, usually measured in seconds). We define:

- $z(x, t)$: surface elevation

- $h(x, t) = z(x, t) - z_b(x)$: river depth at (x, t)
- $A(x, t)$: cross-sectional wetted area
- $Q(x, t)$: flow rate
- $P(x, t)$: wetted perimeter
- $R(x, t) = A(x, t)/P(x, t)$: hydraulic radius
- g : acceleration of gravity (approximately $9.81m/s^2$)

The Saint-Venant equations consist of two main components:

Equation (18) describes mass conservation and equation (19) represents conservation of the linear momentum.

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0 \quad (18)$$

and

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial x} \left(\frac{Q^2}{A} \right) + gA \frac{\partial z}{\partial x} + \frac{n_g^2 Q |Q|}{AR^{4/3}} = 0. \quad (19)$$

The (Yen-Manning) coefficient n_g represents roughness [9].

Typically, given the inlet discharge $Q(x_{min}, t)$ for $t \in [t_{min}, t_{max}]$, initial conditions $A(x, t_{min})$ and $Q(x, t_{min})$, and the coefficient n_g , numerical methods compute the values of $Q(x, t)$ and $A(x, t)$ at a set of discrete “grid points” (x_ν, t_ν) . Unfortunately, the Yen-Manning coefficient n_g is often unknown for natural channels and can vary with position x and even time t . See [9].

In this research, we assume that the Yen-Manning coefficient n_g varies with position x , and that a set of elevation observations $z(x_\nu, t_\nu)$ are available at n_{obs} distinct locations and times, indexed by $\nu = 1, \dots, n_{obs}$. Consequently, the spatial distribution of $n_g(x)$, discretized over a suitable grid, becomes an unknown component of our problem. Furthermore, to ensure physically plausible results, we seek a minimal variation function $n_g(x)$ that minimizes spatial oscillations while fitting the solution of the Saint-Venant equations to the observations.

Considering a grid with n_{grid} points, distributed as $x_{min} \leq x_1, \dots, x_{n_{grid}} \leq x_{max}$, the fitting problem can be formulated in different ways. For example, in (20) we minimize the sum of squares or errors and in (21) we minimize the sum of absolute values. Although we have implemented both possibilities, for the sake of brevity, we will present only the results using (20). In qualitative terms, the results using (21) are very similar.

$$\text{Minimize } \sum_{i=1}^{n_{grid}} (n_g(x_i) - \bar{n})^2 \quad (20)$$

or

$$\text{Minimize } \sum_{i=1}^{n_{grid}} |n_g(x_i) - \bar{n}| \quad (21)$$

subject to the constraint that the simulated elevations match the observed values:

$$z(x_\nu, t_\nu, n_g) = z_{obs}(x_\nu, t_\nu), \quad \nu = 1, \dots, n_{obs}. \quad (22)$$

The unknowns of this problem are $n_g(x_i), i = 1, \dots, n_{grid}$ and $\bar{n}$ and we have n_{obs} equality constraints. Moreover, additional constraints must be introduced:

$$n_g^{max} \geq n_g(x_i) \geq 0 \text{ for all } i = 1, n_{grid}. \quad (23)$$

So far, (20)–(23) define a constrained optimization problem that has the form of (1). The unknowns are $n_g(i), i = 1, \dots, n_{grid}$ and $\bar{n}$. The constraints are given by (22), and the set Ω is defined by (23). The number of unknowns is $n_{grid} + 1$ and the number of constraints (excluding Ω) is n_{obs} .

We found it sensible to tackle this problem using Algorithm 2.1, where the iterates x^k can be computed via different box-constrained derivative-free minimization techniques. Specifically, we are interested in comparing the traditional BOBYQA algorithm [48], introduced and implemented by M. J. D. Powell, with its modified version proposed by Ragonneau and Zhang [49]. For simplicity, the BOBYQA algorithm coded by Powell will be called BOBYQA here, whereas the BOBYQA modification [49] will be called PRIMA, which is the denomination adopted in [49]. BOBYQA has demonstrated excellent performance in numerical studies [50] even before its updating. The modifications of [49] address minor flaws and enhance BOBYQA’s performance, particularly when handling cases where the objective function is ill-defined. This situation frequently arises when the function involves the numerical solution of a Partial Differential Equation (PDE), as in our case.

4 Numerical results

For the numerical experiments, boundary data and initial conditions were used for approximately 3.3 kilometers of the East Fork River, located in the state of Wyoming, USA, as available in [41]. These data were collected every 12 hours between May 17 and June 17 at different points along the river, as illustrated in Figure 1. Additionally, elevation data ($z(x, t)$) at $x = 751m$ and $x = 3256m$, collected every 4 hours starting from May 20 (three days after the initial conditions), were incorporated into constraint (22) of the model, totaling 168 constraints.

To limit the Manning coefficient in constraint (23), the condition $0 \leq n_g(x_i) \leq 1$ was adopted. The differential equations were solved using the explicit finite difference method with artificial diffusion of 0.99 [32, 44]. This method was chosen for its computational simplicity, considering that the equations needed to be solved many times. The spatial discretization was performed with $n_{grid} = 101$ equally spaced points, using a time step of $\Delta t = 1s$. In this type of problem, depending on the considered coefficients, it is common to obtain undefined results (NaN). To circumvent this issue, a penalty of 1×10^{27} was applied to the objective function at these points.

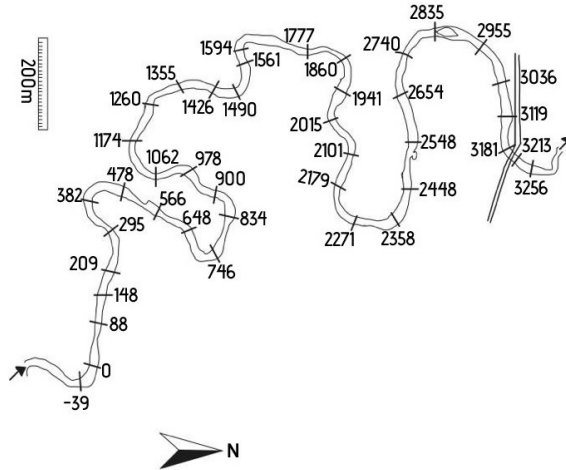


Figure 1: Illustration of the 3.3 km stretch of the East Fork River considered in the model, with markings of the data collection stations used for boundary and initial conditions, recorded between May 17 and June 17, 1977, as available in [41]. Image modified from [6].

To minimize the subproblems generated at Step 2 of Algorithm 2.1, BOBYQA and PRIMA were used. Our codes implementing Algorithm 2.1 and the PRIMA code were written in Fortran 90, while

BOBYQA was written in Fortran 77.

For running Algorithm 2.1 we used the following algorithmic parameters: $\tau = 0.1$, $\gamma = 5$, $\lambda_{min} = -10^3$, $\lambda_{max} = \mu_{max} = 10^3$. Each subproblem was solved using BOBYQA or PRIMA with $\rho_{beg} = 0.05$, $\rho_{end} = 10^{-3}$ and a maximum of 4,000 inner iterations. The initial approximation was $n_g(x_i) = \bar{n} = 0.08$ for every $i = 1, \dots, n_{grid}$. Both BOBYQA and PRIMA used $2n_{grid} + 2$ points for their quadratic interpolations. The feasibility with respect to the constraints (22) is given the “Root Mean-Square Deviation” RMSD defined by

$$\text{RMSD} = \sqrt{\frac{1}{n_{obs}} \sum_{\nu=1}^{n_{obs}} (z(x_\nu, t_\nu, n_g) - z_{obs}(x_\nu, t_\nu))^2}.$$

Note that in this type of problems we do not expect, in general, to obtain feasible points, which may not exist at all. Therefore, we decided to stop the execution of Algorithm 2.1 when Algorithm 2.1 performed 5 outer iterations. As a whole, PRIMA used 2,630 evaluations and obtained RMSD equal to 0.060 m, while BOBYQA required 4,196 evaluations obtaining RMSD equal to 0.053 m.

The obtained results are presented in the form of hydrograms in Figure 2.

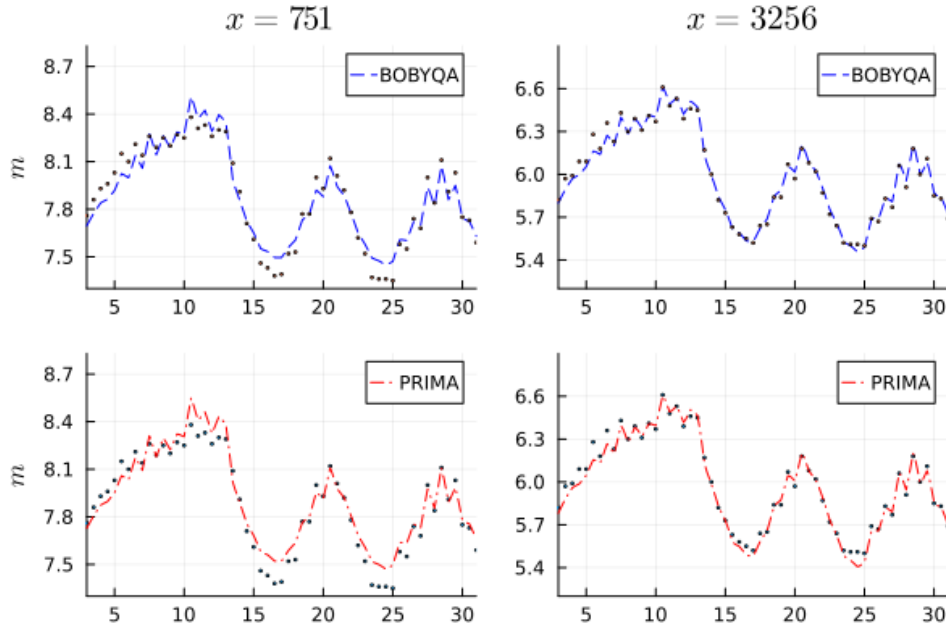


Figure 2: Hydrograms at positions $x = 751$ meters and $x = 3256$ meters comparing the results of (20) using Algorithm 2.1 with BOBYQA or PRIMA for handling subproblems. Observed data are represented by circular markers. The vertical axis represents the elevation $z(x, t)$ in meters, while the horizontal axis corresponds to time t in days.

In Figure 4 we exhibit the evolution of BOBYQA and PRIMA showing the actual RMSD as a function of the number of functional evaluations. Note that, at the beginning, BOBYQA obtain smaller functional values than PRIMA but, finally, the RMSD obtained by both methods are practically the same. For completeness, we include Figure 5, with the same information of Figure 2 but using the merit function (21).

It is interesting to ask whether the results exhibited in Figure 2 are meaningfully better than those obtained assuming a constant or partially constant Manning coefficient. To investigate this, we performed additional experiments with coarser discretizations of $n_g(x)$. Specifically, we considered discretizations with 2 to n_{grid} points, in increasing order. The results showed that the RMSD error with discretizations

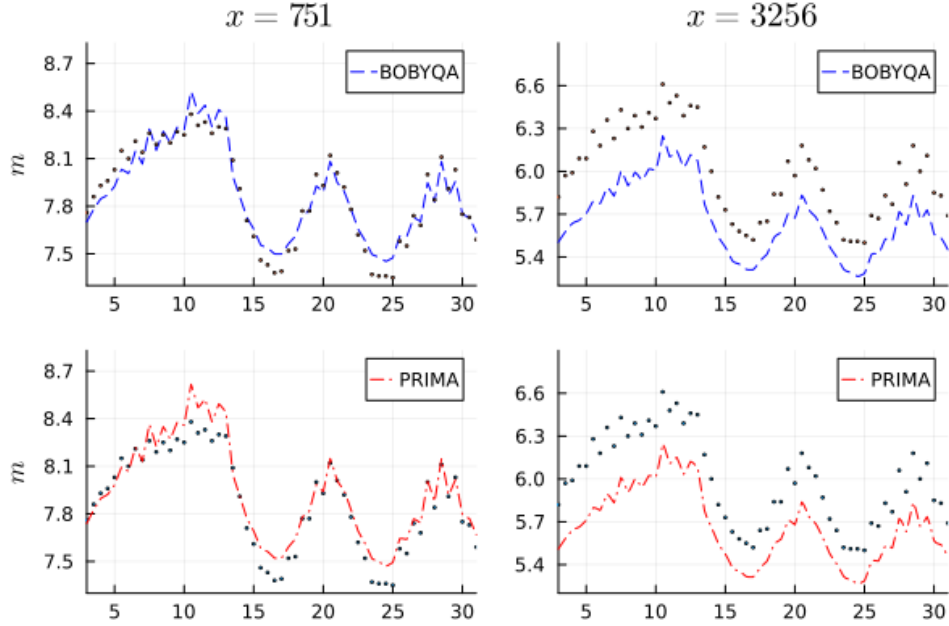


Figure 3: With $n_{grid} = 2$ we observe poor recovery of hydrograms at positions $x = 751$ meters and $x = 3256$ meters. Observed data are represented by circular markers. The vertical axis represents the elevation $z(x, t)$ in meters, while the horizontal axis corresponds to time t in days.

of fewer than 10 points was larger than 0.16 (16 cm), which is substantially larger than the RMSD errors obtained with the full discretization, which were around 6 cm. See Figure 3.

5 Conclusions

We presented a new version of the Safeguarded Augmented Lagrangian method without derivatives. This formulation appears to be well-suited for cases where Derivative-Free methods are used to solve the Augmented Lagrangian subproblems. This is because Derivative-Free methods typically declare convergence when the current iterate exhibits a smaller functional value than its neighbors. To address this, we introduced the auxiliary set N_k , which comprises points with larger functional values than x^k . Consequently, the neighbors used to terminate subproblem solver executions form a subset of N_k . As a result, arguments relying on the differentiability of the objective function or constraints are not employed at all.

The new method is applied to a novel formulation of the problem of estimating Manning coefficients of natural rivers based on the Saint-Venant equations. The novelty lies in the fact that we acknowledge that, in nature, Manning coefficients are not constant, but it is sensible to impose a condition of minimal variation. This formulation gives rise to a constrained optimization problem, whose solution is obtained using the Safeguarded Augmented Lagrangian Method, with the classical BOBYQA algorithm of Powell and its updated version, PRIMA. In our problem, the differences between BOBYQA and PRIMA were not significant.

Augmented Lagrangian methods naturally benefit from advances in box-constrained and linearly-constrained optimization. Since the publication of [17] in 2009, significant progress has been made in unconstrained, box-constrained, and linearly constrained optimization, as seen in works such as [3, 23, 43, 24, 45, 46], among others. The Augmented Lagrangian approach is particularly well-positioned to leverage these advances, providing a versatile framework for constrained optimization. Notably, the integration of recent techniques, such as full-low evaluation [7, 53] and probabilistic descent [27, 28,

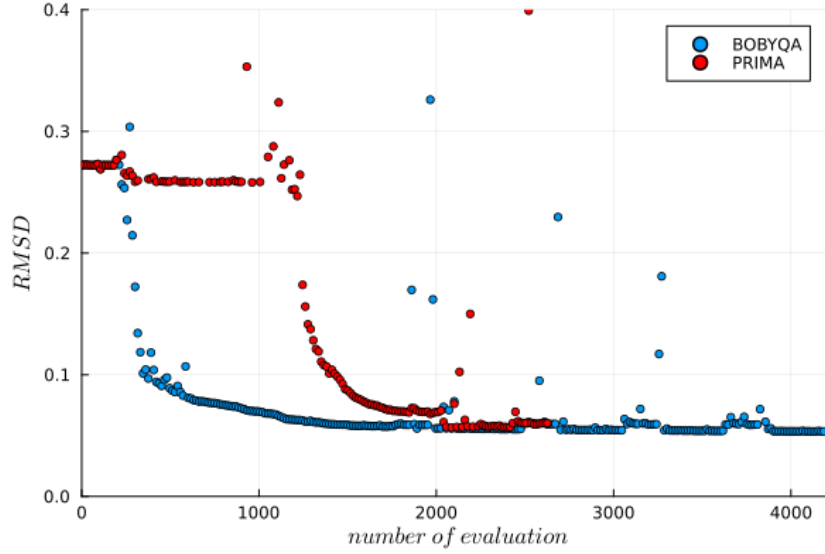


Figure 4: BOBYQA versus PRIMA: Evolution of the RMSD (error in the recovery of elevation observations) of both methods as a function of the number of evaluations. Blue circles correspond to BOBYQA and red circles correspond to PRIMA.

51], into the Augmented Lagrangian framework warrants further exploration and testing for solving subproblems. Furthermore, the increasing tractability of complex subproblem structures, made possible by recent research [8, 18, 20], presents opportunities for developing practical and applicable software by incorporating these subproblems into Augmented Lagrangian frameworks. In particular, multiobjective and equilibrium problems [1, 19, 42] with intricate social constraints represent a promising application area, deserving special attention and further investigation.

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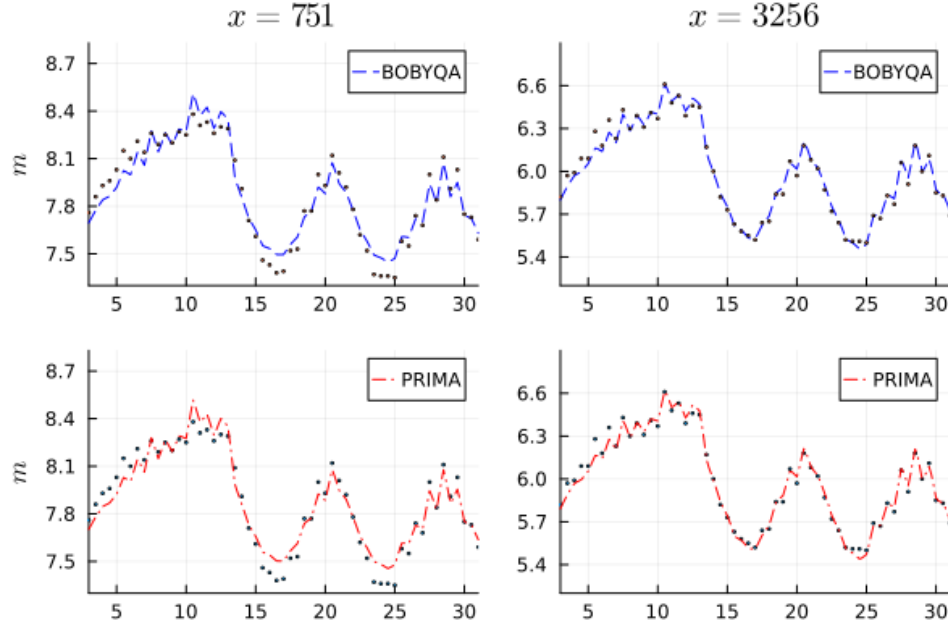


Figure 5: Hydrograms at positions $x = 751$ meters and $x = 3256$ meters comparing the results of (21) using Algorithm 2.1 with BOBYQA or PRIMA for handling subproblems. Observed data are represented by circular markers. The vertical axis represents the elevation $z(x, t)$ in meters, while the horizontal axis corresponds to time t in days.

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